

Discrete-Price-Grid Strict- $h = 0$ Equilibrium

A proposal to break the Grossman-Stiglitz degeneracy

THE PROBLEM (recap):

Under binary payoff, the strict- $h = 0$ operator Φ_0 admits the fully revealing $P_{FR}(u) = \sigma(\tau \sum_k u_k)$ as its unique fixed

point. Conditioning on the level set $\{P_{FR} = p\}$ (a measure-zero hyperplane) gives every trader $\mu_k = p$ pointwise, so no one trades and self-consistency holds trivially. The partially-revealing equilibrium found by the kernel- $h > 0$ operator does not satisfy any natural strict- $h = 0$ fixed-point equation (verified empirically).

THE NEW PROPOSAL (user's idea):

Force the price function to take values on a FIXED DISCRETE GRID:

$$P : \mathbb{R}^3 \rightarrow \{p_1, p_2, \dots, p_M\}$$

with $p_1 < p_2 < \dots < p_M$ prescribed in $(0, 1)$.

Then the conditioning event on observing $P = p_m$ is the PARTITION CELL

$$\Pi_m = \{u \in \mathbb{R}^3 : P(u) = p_m\},$$

a positive-measure subset of $\mathbb{R}^3$, NOT a measure-zero level surface.

CONSEQUENCES:

1. Bayes is classical: $\mu_k(u_k, \Pi_m) = \Pr(v = 1 \mid u_k, u \in \Pi_m)$, well-defined as a conditional probability with positive Bayes denominator.
2. The GS sufficient-statistic identity that pinned $\mu_k = p$ at the FR level set NO LONGER HOLDS on the positive-measure region. There is "room" for $\mu_k \neq p_m$ within each Π_m .
3. Equilibrium = a PARTITION $\{\Pi_1, \dots, \Pi_M\}$ of $\mathbb{R}^3$ such that, for every $u \in \Pi_m$, the CRRA clearing price (given the partition-induced posteriors) equals p_m .

THIS IS A GENUINELY NEW EQUILIBRIUM CONCEPT. It is different from:

- the kernel- $h > 0$ family (which smooths the conditioning event)
- the pointwise strict- $h = 0$ equilibrium (which has only P_{FR})

It corresponds to a real-world feature: prices are quoted on a tick grid (discrete cents), so traders observe a tick-discretized price.

PASS/FAIL:

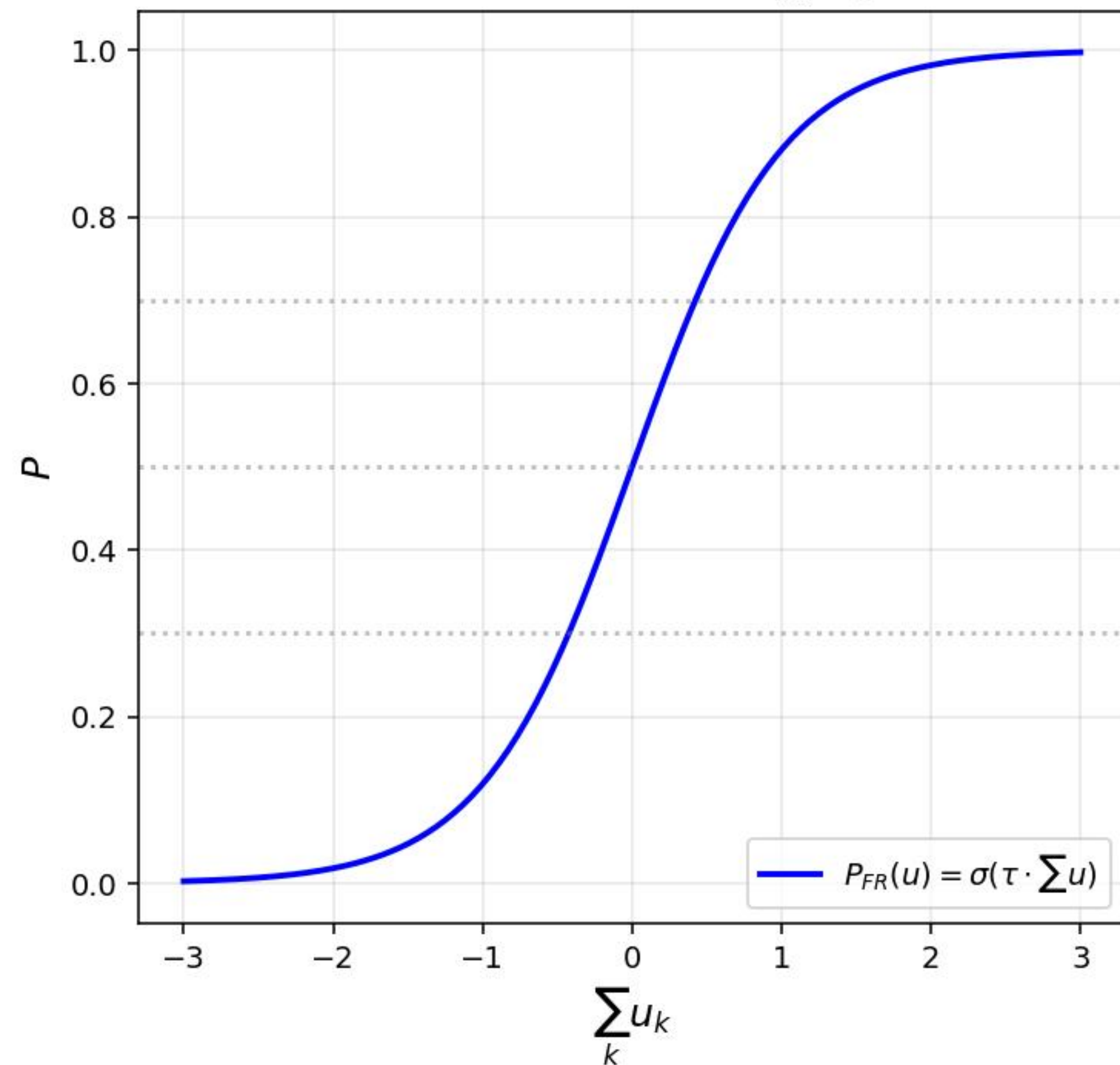
PASS: the solver converges to a partition with Φ -residual zero, positive deficit, distinct from P_{FR} .

FAIL: the only partition that satisfies clearing collapses to the "P_FR-quantized" assignment (each u goes to nearest $P_{FR}(u)$ value on the grid), with degenerate posteriors.

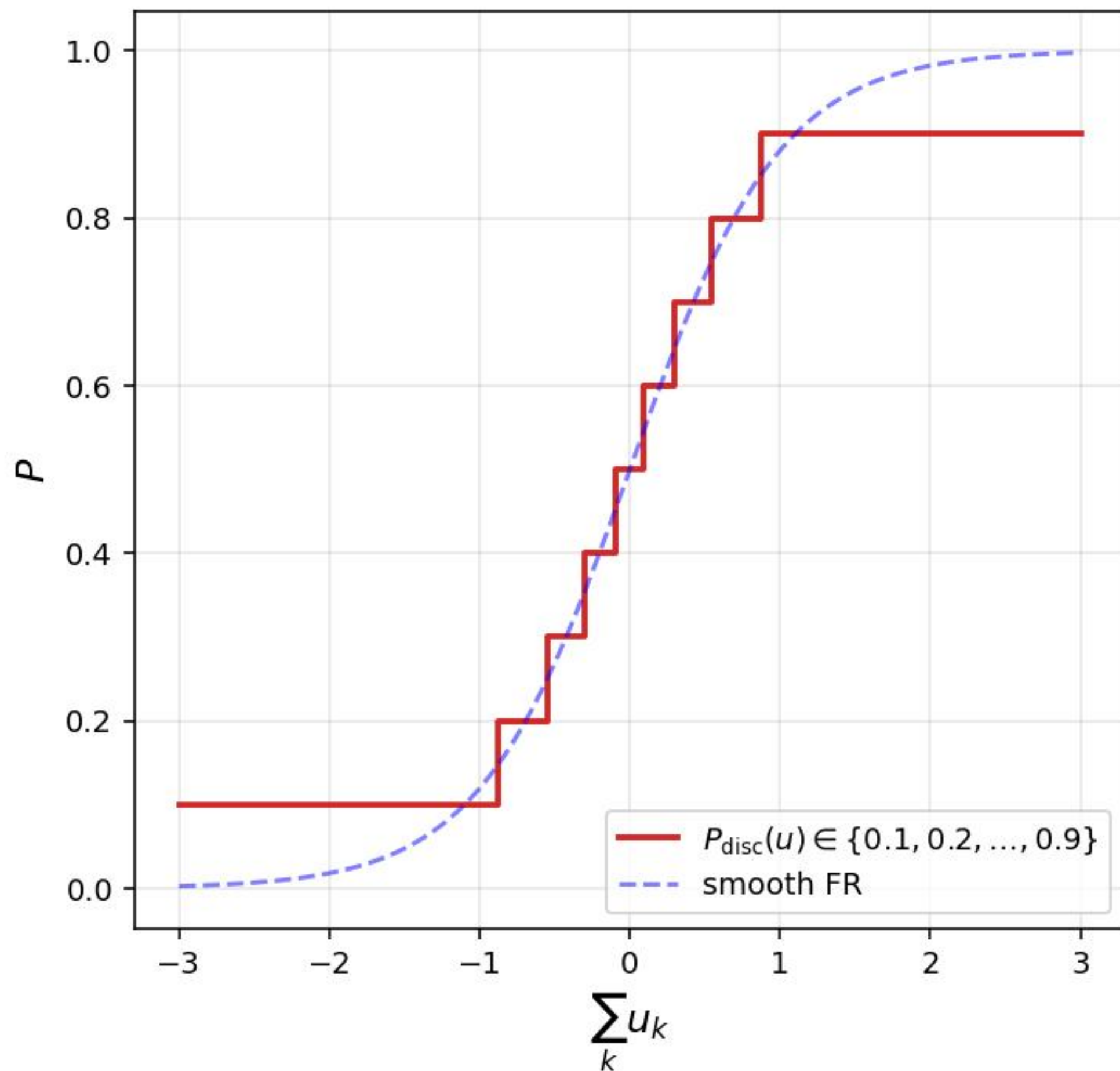
In the FAIL case, even the partition formulation cannot escape P_{FR} -- a much stronger GS degeneracy than the level-set version.

Two representations of the equilibrium price

Smooth (continuous-value) FR price
level sets are measure-zero hyperplanes



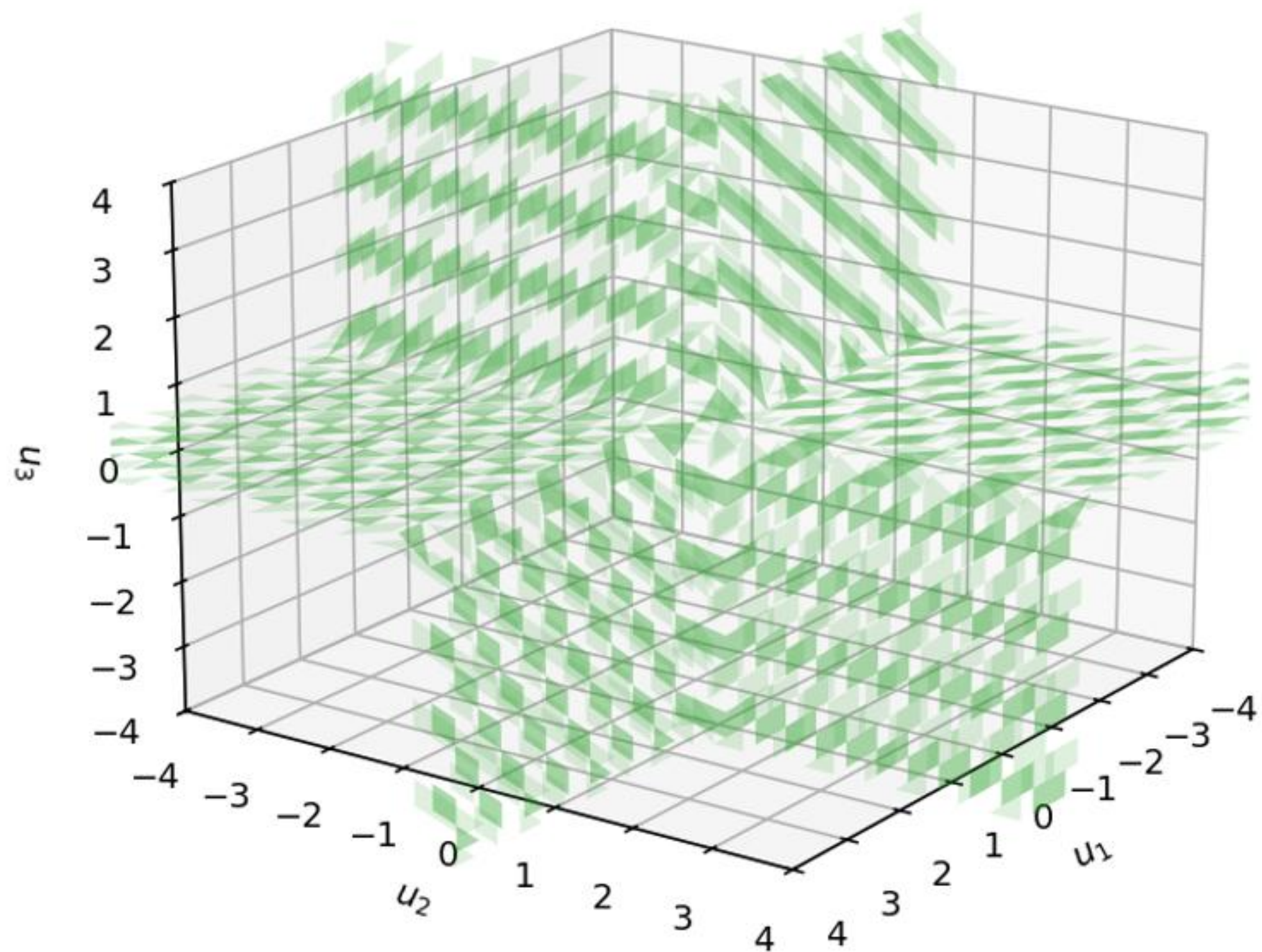
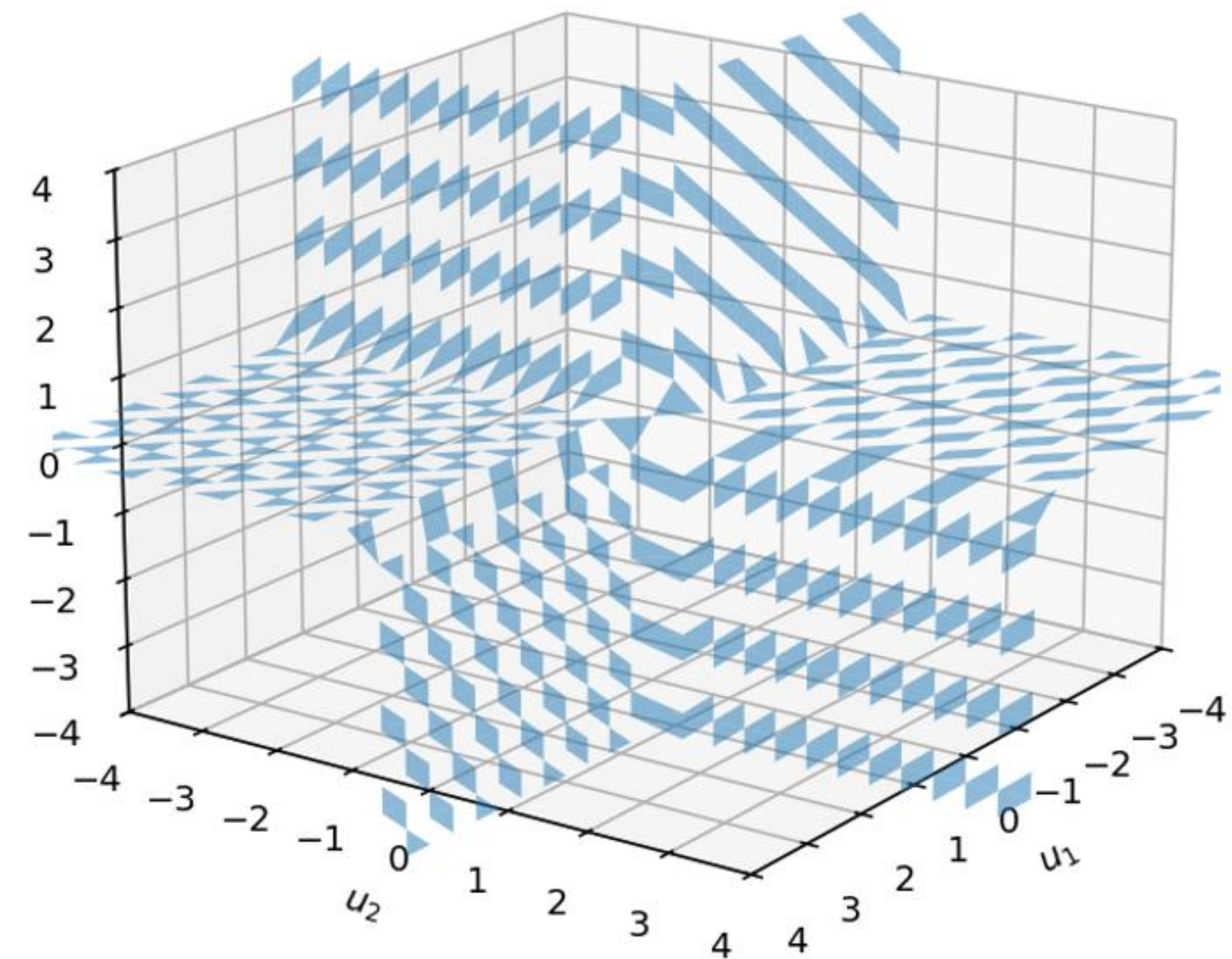
Discrete-price proposal: P takes only finite values
level sets are POSITIVE-MEASURE BANDS



The key difference: conditioning event geometry

Strict- $h = 0$: $\{P = 0.5\}$
2D surface (MEASURE-ZERO)
GS degeneracy: $\mu_k = p$ on this set

Discrete-grid: $\Pi_{0.5} = \{P \in [0.4, 0.6]\}$
3D BAND (POSITIVE-MEASURE)
Bayes classical, no GS degeneracy



Bayes posterior: measure-zero vs positive-measure conditioning

STRICT $h = 0$ (measure-zero level set):

$$\mu_k(u_k, p) = \frac{f_1(u_k) \cdot A_1(p, u_k)}{f_0(u_k) \cdot A_0(p, u_k) + f_1(u_k) \cdot A_1(p, u_k)}$$

where

$$A_v(p, u_k) = \int_{\{P=p\} \cap \{u_k \text{ fixed}\}} f_v(u_j) f_v(u_l) d\sigma$$

PROBLEM: at $P = P_{FR}$, the constraint $\{P = p\}$ pins $\sum_k u_k$ exactly. The Gaussian integrals close in closed form, every trader's posterior collapses to $\mu_k = p$, CRR clearing returns p trivially. P_{FR} is an exact fixed point but the equilibrium is degenerate.

DISCRETE-PRICE GRID (positive-measure partition cell):

$$\mu_k(u_k, \Pi_m) = \frac{\int_{\Pi_m \cap \{u_k \text{ fixed}\}} f_1(u_j) f_1(u_l) du}{\int_{\Pi_m \cap \{u_k \text{ fixed}\}} f_0(u_j) f_0(u_l) du + (\text{symmetric})}$$

The conditioning set $\Pi_m \cap \{u_k \text{ fixed}\}$ is a 2D region (positive measure in $\mathbb{R}^2$), not a 1D curve. Standard Bayes. The Gaussian-closed-form identity that pinned $\mu_k = p$ NO LONGER APPLIES: the integration is over a 2D region, not a 1D level set.

KEY OBSERVATION:

As the price grid is refined ($M \rightarrow \infty$), the partition cells Π_m shrink toward the level surfaces $\{P = p\}$. The Bayes posteriors approach the strict- $h = 0$ posteriors. But for any FINITE M , the equilibrium concept is genuinely different from the strict pointwise version.

For the binary-payoff GS degeneracy, the issue is specifically the MEASURE-ZERO conditioning -- any positive-measure conditioning (whether by kernel smoothing OR by discrete-price quantization) should break the degeneracy.

ECONOMIC INTERPRETATION:

A discrete price grid is what we observe in real markets: prices are quoted in ticks (e.g., \$0.01 increments). An equilibrium under tick-discretized prices is the natural object; the continuous-price abstraction is the model assumption that creates the GS degeneracy under binary payoff.

Solver algorithm

STATE:

Discrete-price grid $p_1 < p_2 < \dots < p_M$ (fixed).

Partition assignment $a: \{1, \dots, G^3\} \rightarrow \{1, \dots, M\}$
(which cube cell goes to which price level).

EQUILIBRIUM CONDITIONS:

For every cube cell c with assignment $a(c) = m$ and signal vector $u^{(c)}$:

1. Posteriors $\mu_k = \mu_k(u_k^{(c)}, \Pi_m)$ via Bayes on the partition.
2. CRRA clearing: $p^{\text{clear}}(\mu_1, \mu_2, \mu_3; \gamma) = p_{a(c)}$.

TWO SOLVER PATHS:

Path A (hard K-means style):

Iterate:

- i. For each cell c : compute current μ_k , then p^{clear} .
- ii. Reassign $a(c) := \arg \min_m |p_m - p^{\text{clear}}(c)|$.
- iii. Stop when assignment stops changing.

Fixed point: partition matches clearing prices.

Discrete dynamics; may oscillate; easy to monitor.

Path B (soft-max relaxation):

Replace hard assignment by soft weights

$$w_m(c) = \text{softmax}_m(\text{logit}(p_m) - \text{logit}(p^{\text{clear}}(c))/T)$$

with temperature $T > 0$. The price field is

$$P(c) = \sum_m p_m w_m(c).$$

Solve $P(c) = p^{\text{clear}}(c)$ as a continuous fixed-point problem
via `scipy.optimize.least_squares`. Anneal $T \rightarrow 0$.

INITIAL GUESS:

Quantize the kernel- $h > 0$ certified solution P_h :

$$a^{(0)}(c) := \arg \min_m |p_m - P_h^*(c)|.$$

This places the kernel solution on the discrete grid.

PASS CRITERION:

Converged partition with $\max_c |p_{a(c)} - p^{\text{clear}}(c)| < \text{tol}$,

where tol should be smaller than half the grid spacing

$$\min_m |p_m - p_{m+1}|/2.$$

FAIL CRITERION:

Iteration converges to the "P_FR-quantized" partition with all

$\mu_k = p_m$ exactly (degenerate posteriors, same as continuous P_FR).

Why a non-trivial discrete-price fixed point should exist

INFORMAL ARGUMENT:

1. The kernel- $h > 0$ family has a unique nontrivial fixed point P_h^* for every $h > 0$, with positive deficit. This is established by the existence theorem of the main paper.
2. The kernel smoothing replaces the indicator $\mathbf{1}\{P(u) = p\}$ by a Gaussian $K_h(P - p)$ supported on $[p - h, p + h]$. This is equivalent to averaging over a band of width $\sim h$.
3. The discrete-price quantization replaces $\mathbf{1}\{P(u) = p\}$ by $\mathbf{1}\{P(u) \in [p_m - \delta_m, p_m + \delta_m]\}$ where the bands are the partition cells Π_m . This is a HARD-EDGED averaging over an explicit partition.
4. Both modifications turn the measure-zero conditioning event into a positive-measure event. By continuity:
 P_h^* exists for all $h > 0$ (proven)
 $\Rightarrow$ discrete-grid fixed point should exist for any partition that "resembles" a kernel-induced one.
5. The natural candidate:
Discretize the kernel solution P_h^* on a price grid with spacing $\sim h$. The resulting partition has Π_m similar in size to the kernel's effective conditioning band.
6. If the soft-max relaxation converges, the limit $T \rightarrow 0$ gives a hard partition that approximately satisfies the discrete-grid fixed-point equation.

POTENTIAL OBSTRUCTION:

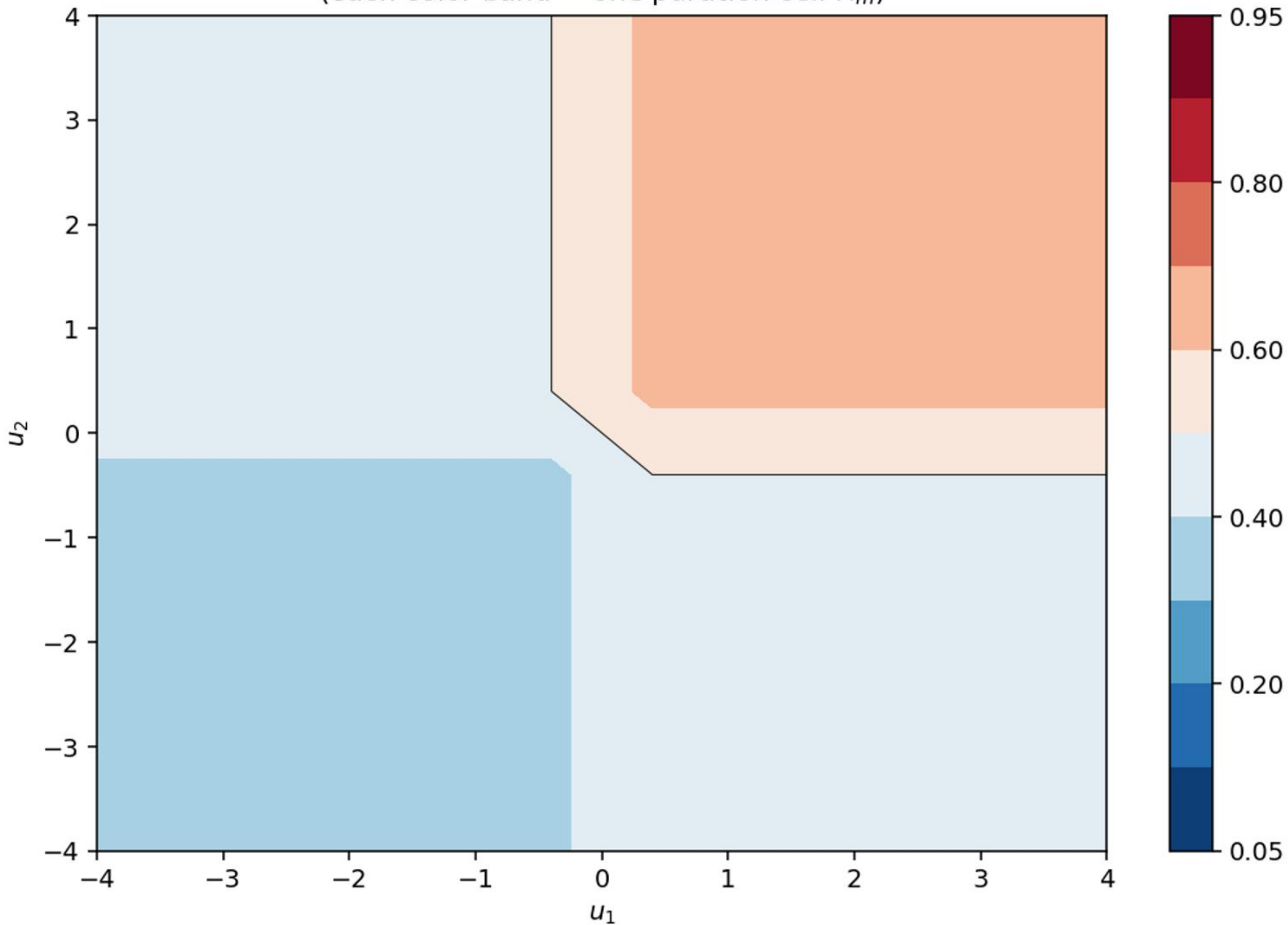
The hard-K-means iteration may have NO fixed point. This happens if the clearing price under any candidate partition always pulls cells AWAY from their assigned level. In particular, if all clearing prices cluster near 0.5 (regardless of partition), then the "P_FR-quantized" partition is the unique attractor, even at the partition level. This would be a STRONGER GS degeneracy.

EITHER OUTCOME IS INFORMATIVE:

PASS: we have a strict- $h = 0$ partition equilibrium with positive deficit, breaking the degeneracy that the continuous-P version had. The paper gains a new equilibrium concept and a real-world interpretation (tick-discretized prices).

FAIL: the GS degeneracy is even stronger than previously thought -- even discrete-price conditioning collapses to FR. This would be a striking impossibility result.

Initial guess: certified kernel $P_h^*(u_1, u_2, u_3 = 0)$ quantized to
the discrete price grid $\{0.05, 0.1, 0.2, 0.3, \dots, 0.95\}$
(each color band = one partition cell Π_m)



Implementation roadmap

ROADMAP (this PDF will be followed by the actual implementation):

STAGE 1: hard K-means iteration at $G=8$

- Compute kernel-certified P at $G=8$ (~5 seconds)
- Quantize to discrete grid $M = 8, 16$
- Iterate: clearing price $\rightarrow$ reassign $\rightarrow$ repeat
- Monitor: assignment change rate, $\max |p_{\text{clear}} - p_{\text{assigned}}|$
- Expected wall: minutes per (G, M) combination

STAGE 2: soft-max relaxation at $G=8$, then $G=11$

- `scipy.optimize.least_squares` on soft assignment weights
- Temperature anneal $T = 0.1 \rightarrow 0.01$
- Final hard assignment via `argmax` of soft weights
- Verify hard partition satisfies the equilibrium

STAGE 3: refinement test

- Refine M ($8 \rightarrow 16 \rightarrow 32$ levels) and check whether deficit converges to the kernel-h limit
- Refine G ($8 \rightarrow 11 \rightarrow 15$) and check stability

STAGE 4: documentation

- Build "discrete-price equilibrium" report with all results
- Update main paper with new equilibrium concept
- Compare with kernel-h limit empirically

PASS/FAIL DELIVERABLE:

At end of Stage 1-2, definitive answer to "does a non-trivial discrete-price strict- $h=0$ equilibrium exist for this model?"